



1.8.6 Wrap-Up: Cool Down

1. Explain a way that you can determine if a solution is correct for a given inequality.
2. How do you know if the circle should be open or closed when graphing an inequality?



Unit 1, Lesson 9: Solving Inequalities – Part 1



Benchmark

MA.8.AR.2.2 Solve two-step linear inequalities in one variable and represent solutions algebraically and graphically.



Learning Target

- ☐ I can solve inequalities and represent the solutions on a number line.



1.9.1 Warm-Up: Bell Work

1. Solve and graph each inequality.

a. $35 < 5 - 3x$



b. $6.5 - \frac{3}{4}n > 12.5$



1.9.2 Guided Instruction: Solving Multi-Step Inequalities

Sadie and Promus each solve the inequality $3(x - 6) \geq 9$.

Sadie's Work	Promus' Work
$3(x - 6) \geq 9$	$3(x - 6) \geq 9$
$3x - 18 \geq 9$	$\frac{3}{3} \quad \frac{9}{3}$
$+18 \quad +18$	$x - 6 \geq 3$
$3x \geq 27$	$+6 \quad +6$
$x \geq 9$	$x \geq 9$

1. Compare and contrast Sadie's and Promus' work.

2. Graph the solution to the inequality $3(x - 6) \geq 9$.



3. Use substitution to verify the solution set.

4. Solve the inequality $56 < 7(7 - x)$ using Sadie's method and Promus' method.

Sadie's Method	Promus' Method
$56 < 7(7 - x)$	$56 < 7(7 - x)$

5. Graph the solution to the inequality $56 < 7(7 - x)$.



6. Use substitution to verify the solution set.

7. Explain whether you prefer Sadie's method or Promus' method.



1.9.3 Guided Practice: Solving Inequalities

1. Solve each inequality and graph the solution set.

a. $6(h + 2) \geq -12$ 

b. $-10 > -\frac{2}{3}(p - 3)$ 

c. $5.6(-4.5 + x) < -36.4$ 

d. $\frac{3}{4} \geq \frac{1}{8}(\frac{4}{5}b - 6)$ 



1.9.4 Collaboration: Error Analysis

- Work with your partner to identify the errors in the table. Circle the mistake, solve the inequality correctly, then graph the solution set.

Inequality	Correction	Graph
$2(3x - 2) > -2$ $6x - 4 > -2$ $+4 \quad +4$ $6x > 2$ $-6 \quad -6$ $x > -4$		
$5 - \frac{y}{3} \geq 3$ $-5 \quad -5$ $(-3) \cdot -\frac{y}{3} \geq -2 \cdot (-3)$ $y \geq -6$		
$2.5 - 4a \geq 10.5$ $-2.5 \quad -2.5$ $4a \geq 8$ $\frac{4}{4} \quad \frac{4}{4}$ $a \geq 2$		
$3(2n - 3) < 0$ $\frac{3}{3} \quad \frac{3}{3}$ $3n - 3 < 0$ $+3 \quad +3$ $\frac{3n}{3} < \frac{3}{3}$ $n < 1$		



1.9.5 Your Turn!

- Solve the inequality $5(x - 1) > -40$ and graph the solution set on a number line.



- Describe how to solve the inequality $32 \geq -4(y - 10)$.



1.9.6 Wrap-Up: Reflection

1. Complete each statement to reflect on today's learning.
 - a. When someone asks me what I did in math today, I can say ...
 - b. The thing that made the most sense to me today was ...
 - c. Something that I still do not understand is ...



Unit 1, Lesson 10: Solving Inequalities – Part 2



Benchmark

MA.8.AR.2.2 Solve two-step linear inequalities in one variable and represent solutions algebraically and graphically.



Learning Target

- ☐ I can solve inequalities and represent the solutions on a number line.